

Hands-on Assignments for Restricting and Sorting Data

Complete the below hands-on assignments before proceeding with the next Topic

Update Completion Status

No.	Cat	Hands-on Assignment	Topics Covered
1	M	Display the Ename,Sal,Comm from Emp table who earn commission and sort the records in descending order of Salary and Comm. Use column's numeric position in the ORDER BY clause	Sorting Data

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```
SELECT e.LAST_NAME AS ENAME,
       e.SALARY AS SAL,
       e.COMMISSION_PCT AS COMM
FROM HR.EMPLOYEES e
WHERE e.COMMISSION_PCT IS NOT NULL
ORDER BY 2 DESC, 3 DESC;
```

0.082 seconds

Script Output

Task completed in 0.082 seconds

ENAME	SAL	COMM
Singh	14000	.4
Partners	13500	.3
Errazuriz	12000	.3
Ozer	11500	.25
Cambrault	11000	.3
Abel	11000	.3
Vishney	10500	.25
Zlotkey	10500	.2
King	10000	.35
Tucker	10000	.3
Bloom	10000	.2

2	M	The HR department needs a query to display all unique job codes from the EMP table.	Functions
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Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

0.155 seconds

Worksheet Query Builder

```
SELECT DISTINCT JOB_ID
FROM HR.EMPLOYEES
ORDER BY JOB_ID;
```

Script Output x

Task completed in 0.155 seconds

JOB_ID

AC_ACCOUNT
AC_MGR
AD_ASST
AD PRES
AD_VP
FI_ACCOUNT
FI_MGR
HR_REP
IT_PROG
MK_MAN
MK_REP

JOB_ID

PR_REP
PU_CLERK
PU_MAN
SA_MAN
SA_REP
SH_CLERK
ST_CLERK
ST_MAN

19 rows selected.

3

M

The HR department wants more descriptive column headings for its report on employees. Name the column headings Emp #, Employee, Job, and Hire Date, respectively by giving Column Alias.

Columns Alias

Oracle SQL Developer : Oracle

Find Database Object

Oracle

0.19599999 seconds

Worksheet Query Builder

```
SELECT EMPLOYEE_ID AS "Emp #",
       LAST_NAME AS "Employee",
       JOB_ID AS "Job",
       HIRE_DATE AS "Hire Date"
FROM HR.EMPLOYEES;
```

Script Output

Task completed in 0.196 seconds

Emp #	Employee	Job	Hire Dat
100	King	AD_PRES	17-06-13
101	Yang	AD_VP	21-09-15
102	Garcia	AD_VP	13-01-11
103	James	IT_PROG	03-01-16
104	Miller	IT_PROG	21-05-17
105	Williams	IT_PROG	25-06-15
106	Jackson	IT_PROG	05-02-16
107	Nguyen	IT_PROG	07-02-17
108	Gruenberg	FI_MGR	17-08-12
109	Faviet	FI_ACCOUNT	16-08-12
110	Chen	FI_ACCOUNT	28-09-15

Emp #	Employee	Job	Hire Dat
111	Sciarra	FI_ACCOUNT	30-09-15
112	Urman	FI_ACCOUNT	07-03-16
113	Popp	FI_ACCOUNT	07-12-17
114	Li	PU_MAN	07-12-12
115	Khoo	PU_CLERK	18-05-13
116	Baida	PU_CLERK	24-12-15
117	Tobias	PU_CLERK	24-07-15
118	Himuro	PU_CLERK	15-11-16
119	Colmenares	PU_CLERK	10-08-17
120	Weiss	ST_MAN	18-07-14
121	Frapp	ST_MAN	10-04-15

Emp #	Employee	Job	Hire Dat
122	Kaufling	ST_MAN	01-05-13

4

M

The HR department has requested a report of all employees and their job IDs. Display the last name concatenated with the job ID (separated by a comma and space) and name the column Employee and Title by giving Column Alias.

Operators

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Oracle hr_install.sql 0.147 seconds

Worksheet Query Builder

```
SELECT LAST_NAME || ', ' || JOB_ID AS "Employee and Title"
FROM HR.EMPLOYEES;
```

Script Output x

Task completed in 0.147 seconds

Employee and Title

Abel, SA_REP
 Ande, SA_REP
 Atkinson, ST_CLERK
 Baida, PU_CLERK
 Banda, SA_REP
 Bates, SA_REP
 Bell, SH_CLERK
 Bernstein, SA_REP
 Bissot, ST_CLERK
 Bloom, SA_REP
 Brown, PR_REP

Employee and Title

Bull, SH_CLERK
 Cabrio, SH_CLERK
 Cambrault, SA_MAN
 Cambrault, SA_REP
 Chen, FI_ACCOUNT
 Chung, SH_CLERK
 Colmenares, PU_CLERK
 Davies, ST_CLERK
 Davis, MK_REP
 Dellinger, SH_CLERK
 Dilly, SH_CLERK

Employee and Title

Doran, SA_REP
 Errazuriz, SA_MAN
 Everett, SH CLERK

5

M

To familiarize yourself with the data in the EMP table, create a query to display all the data from that table. Separate each column output by a comma. ENAME, JOB, HIREDATE, MGR. Name the column title THE_OUTPUT.

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```
SELECT EMPLOYEE_ID  
FIRST_NAME  
LAST_NAME  
EMAIL  
PHONE_NUMBER  
HIRE_DATE  
JOB_ID  
SALARY  
COMMISSION_PCT  
MANAGER_ID  
DEPARTMENT_ID AS THE_OUTPUT  
FROM HR.EMPLOYEES;
```

Script Output

Task completed in 0.151 seconds

THE_OUTPUT

100	Steven	King	SKING	1.515.555.0100	17-06-13	AD_PRES	24000	,,90
101	Neena	Yang	NYANG	1.515.555.0101	21-09-15	AD_VP	17000	,100,90
102	Lex	Garcia	LGARCIA	1.515.555.0102	13-01-11	AD_VP	17000	,100,90
103	Alexander	James	AJAMES	1.590.555.0103	03-01-16	IT_PROG	9000	,102,60
104	Bruce	Miller	BMILLER	1.590.555.0104	21-05-17	IT_PROG	6000	,103,60
105	David	Williams	DWILLIAMS	1.590.555.0105	25-06-15	IT_PROG	4800	,103,60
106	Valli	Jackson	VJACKSON	1.590.555.0106	05-02-16	IT_PROG	4800	,103,60
107	Diana	Nguyen	DNGUYEN	1.590.555.0107	07-02-17	IT_PROG	4200	,103,60
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	17-08-12	FI_MGR	12008	,101,100
109	Daniel	Faviet	DFAVIET	1.515.555.0109	16-08-12	FI_ACCOUNT	9000	,108,100
110	John	Chen	JCHEN	1.515.555.0110	28-09-15	FI_ACCOUNT	8200	,108,100
111	Ismael	Sciarra	ISCIARRA	1.515.555.0111	30-09-15	FI_ACCOUNT	7700	,108,100
112	Jose Manuel	Urman	JMURMAN	1.515.555.0112	07-03-16	FI_ACCOUNT	7800	,108,100
113	Luis	Popp	LPOPP	1.515.555.0113	07-12-17	FI_ACCOUNT	6900	,108,100
114	Den	Li	DLI	1.515.555.0114	07-12-12	PU_MAN	11000	,100,30
115	Alexander	Khoo	AKHOO	1.515.555.0115	18-05-13	PU_CLERK	3100	,114,30

6

M

Create a report to display the ename, job , and Hiredate for the employees name is SCOTT or TURNER. Order the query in ascending order by hiredate.

Sorting Data

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns

Worksheet

Query Builder

```
SELECT LAST_NAME AS NAME,  
JOB_ID AS JOB,  
HIRE_DATE AS HIREDATE  
FROM HR.EMPLOYEES  
WHERE UPPER(LAST_NAME) IN ('SCOTT', 'TURNER')  
ORDER BY HIRE_DATE ASC;
```

7

M

Display the ename and department number of all employees in departments 20 or 30 in ascending alphabetical order by ename.

Sorting Data

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```

SELECT LAST_NAME AS ENAME,
       DEPARTMENT_ID AS DEPTNO
FROM HR.EMPLOYEES
WHERE DEPARTMENT_ID IN (20, 30)
ORDER BY LAST_NAME ASC;
  
```

0.148 seconds

Script Output

Task completed in 0.148 seconds

ENAME	DEPTNO
Baida	30
Colmenares	30
Davis	20
Himuro	30
Khoo	30
Li	30
Martinez	20
Tobias	30

8 rows selected.

9 M The HR department needs a report that displays the last name and hire date for all employees who were hired in 1981

Operators

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns

Worksheet

Query Builder

```

SELECT LAST_NAME,
       HIRE_DATE
FROM HR.EMPLOYEES
WHERE EXTRACT(YEAR FROM HIRE_DATE) = 1981;
  
```

0.101 seconds

8 M Modify the previous query to display the last name and salary of employees who earn between 2000 and 3000 and are in department 20 or 30. Label the columns Employee and Monthly Salary, respectively giving Column Alias.

Sorting Data and Column Alias

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```

SELECT LAST_NAME AS "Employee",
       SALARY AS "Monthly Salary"
FROM HR.EMPLOYEES
WHERE SALARY BETWEEN 2000 AND 3000
      AND DEPARTMENT_ID IN (20, 30)
ORDER BY LAST_NAME;
  
```

Script Output

Task completed in 0.124 seconds

Employee	Monthly Salary
Baida	2900
Colmenares	2500
Himuro	2600
Tobias	2800

10

M

Display the Ename, Sal of employees who earn more than an amount the user specifies after a Prompt.

Using Prompts

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```

SELECT LAST_NAME AS ENAME,
       SALARY AS SAL
FROM HR.EMPLOYEES
WHERE SALARY > &amount;
  
```

Script Output

Task completed in 5.53 seconds

old:SELECT LAST_NAME AS ENAME,
SALARY AS SAL
FROM HR.EMPLOYEES
WHERE SALARY > &amount
new:SELECT LAST_NAME AS ENAME,
SALARY AS SAL
FROM HR.EMPLOYEES
WHERE SALARY > 10000

ENAME	SAL
King	24000
Yang	17000
Garcia	17000
Gruenberg	12008
Li	11000
Singh	14000
Partners	13500
Errazuriz	12000
Cambrault	11000
Zlotkey	10500
Vishney	10500
Ozer	11500
Abel	11000
Martinez	13000
Higgins	12008

11

M

Create a report to display the last name and job title of all employees who do not have a manager.

SQL Operators

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```
SELECT e.LAST_NAME,
       j.JOB_TITLE
FROM HR.EMPLOYEES e
JOIN HR.JOBS j ON e.JOB_ID = j.JOB_ID
WHERE e.MANAGER_ID IS NULL;
```

Script Output

Task completed in 0.152 seconds

LAST_NAME	JOB_TITLE
King	President

12

M

Create a query that prompts the user for Manager ID and generate EMPNO, ENAME, SAL, DEPTNO. The user should have the ability to sort the records on a selected Column.

Using Prompts and Sorting Data

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```
SELECT EMPLOYEE_ID AS EMPNO,
       LAST_NAME AS ENAME,
       SALARY AS SAL,
       DEPARTMENT_ID AS DEPTNO
FROM HR.EMPLOYEES
WHERE MANAGER_ID = &mgr_id
ORDER BY &sort_col;
```

Script Output

Task completed in 5.627 seconds

```
old:SELECT EMPLOYEE_ID AS EMPNO,
       LAST_NAME AS ENAME,
       SALARY AS SAL,
       DEPARTMENT_ID AS DEPTNO
FROM HR.EMPLOYEES
WHERE MANAGER_ID = &mgr_id
ORDER BY &sort_col
new:SELECT EMPLOYEE_ID AS EMPNO,
       LAST_NAME AS ENAME,
       SALARY AS SAL,
       DEPARTMENT_ID AS DEPTNO
FROM HR.EMPLOYEES
WHERE MANAGER_ID = 1
ORDER BY 2
no rows selected
```

13

M

The HR department wants to run reports based on a manager. Create a query that prompts the user for a MGR and generates the empno, ename, salary, and department for that manager's employees. The HR department wants the ability to sort the report on a selected column

Using Prompts

Oracle SQL Developer : Oracle

Find Database Object

Oracle

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```
e.LAST_NAME      AS ENAME,  
e.SALARY         AS SAL,  
d.DEPARTMENT_NAME AS DEPARTMENT  
FROM HR.EMPLOYEES e  
JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID  
WHERE e.MANAGER_ID = &mgr  
ORDER BY &sort_col;
```

Script Output

Task completed in 5.905 seconds

```
old:SELECT e.EMPLOYEE_ID      AS EMPNO,  
       e.LAST_NAME          AS ENAME,  
       e.SALARY             AS SAL,  
       d.DEPARTMENT_NAME    AS DEPARTMENT  
FROM HR.EMPLOYEES e  
JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID  
WHERE e.MANAGER_ID = &mgr  
ORDER BY &sort_col  
new:SELECT e.EMPLOYEE_ID      AS EMPNO,  
       e.LAST_NAME          AS ENAME,  
       e.SALARY             AS SAL,  
       d.DEPARTMENT_NAME    AS DEPARTMENT  
FROM HR.EMPLOYEES e  
JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID  
WHERE e.MANAGER_ID = 2  
ORDER BY 3  
no rows selected
```

14 M Display all employee last names in which the third letter of the name is A SQL Operators

Oracle SQL Developer : C

Find Database Object

Oracle

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages

Worksheet

Query Builder

```
SELECT LAST_NAME  
FROM HR.EMPLOYEES  
WHERE LAST_NAME LIKE '__A%';
```

0.142 seconds

15 M Display the last name of all employees who have both an A and an S in their ename SQL Operators

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

- ☒ All Schemas
- ☒ All Object Types
- ☐ All Identifier Types
- ☐ All Identifier Usages
- ☐ Columns
- ☐ Code
- ☐ All Dependencies

Worksheet

Query Builder

```
SELECT LAST_NAME
FROM HR.EMPLOYEES
WHERE UPPER(LAST_NAME) LIKE '%A%'
AND UPPER(LAST_NAME) LIKE '%S%';
```

Script Output

Task completed in 0.147 seconds

```
LAST_NAME
-----
Atkinson
Bates
Colmenares
Davies
Davis
Jackson
Jacobs
James
Marvins
Matos
Partners

LAST_NAME
-----
Rajs
Sarchand
Sciarra
Sewall
Sullivan
Tobias
Vargas
Walsh
Williams

20 rows selected.
```

16 M Display the Ename, Job, Sal for all employees whose jobs are CLERK and whose salary is in 800 or 950 or 1300. SQL Operators

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

- ☒ All Schemas
- ☒ All Object Types
- ☐ All Identifier Types
- ☐ All Identifier Usages
- ☐ Columns
- ☐ Code
- ☐ All Dependencies

Worksheet

Query Builder

```
SELECT e.LAST_NAME AS ENAME,
       j.JOB_TITLE AS JOB,
       e.SALARY AS SAL
FROM HR.EMPLOYEES e
JOIN HR.JOBS j ON e.JOB_ID = j.JOB_ID
WHERE UPPER(j.JOB_TITLE) LIKE '%CLERK%'
AND e.SALARY IN (800, 950, 1300);
```